

Analysis of Dataset

Overview

This overview describes attributes of the main dataset (`AppReviewData.csv`). In this dataset, there are six columns/features: `appID`, `reviewerName`, `reviewText`, `reviewerRating`, `reviewDate`, and `textAnalytics`. In addition, there are 111143 rows of unique reviews.

Limitations / Considerations:

- There is a possibility that the limited number of features could affect building a robust ML model.

Feature: `appID`

For this feature, there are 492 unique apps that have been reviewed. The range of reviews per app is quite large. The app with the most reviews is `app-2078` with 4,213 reviews, and there are 35 apps with only 1 review (the lowest amount of reviews possible).

Limitations / Considerations:

- `AppID` alone rarely explains the sentiment or content of a review.
- Model may skew results on apps with higher review amounts.

Feature: `reviewerName`

With an initial query to find the unique count of reviewers, there was a count of 79,372 unique users submitting reviews. However, when looking at a value count, about 20,000 of those users were listed as “A Google User”. These indicate guest user reviews, and so they should be counted as individual users to get an accurate account of how many people would be submitting reviews. So with this in mind, the total number of unique Users submitting reviews is around 100,000. Further, the user with the highest amount of reviews is John Smith with 26, followed by 6 users with at least 10 reviews.

Limitations / Considerations:

- Name alone rarely explains the sentiment or content of a review.
- The ML model may learn to associate specific names with particular outcomes (e.g., "John Smith always gives positive reviews") rather than learning underlying patterns, resulting in poor performance on new data.
- The model will struggle to make predictions for new, unseen names, rendering the feature useless for new user inference.

Feature: reviewText

There are 100,936 unique written reviews. Most of the reviews, 98,830, are only seen once in the entire dataset, typically indicating these are at least a sentence long or contain a unique variable of text. The rest are simple and easily repeatable strings of text, such as one-word responses, such as “Good”, “Ok”, or “Great!”. Some of these simple ones are further separated by capitalization inconsistencies, so “Nice” and “nice” are considered different variables. Would likely want to capitalize all values to remove these discrepancies.

Limitations / Considerations:

- Subjectivity of scores could be hard to account for. A 3-star review might be "good" to one user and "bad" to another.
- Mixed sentiment in a review could confuse the ML model.
- Sarcasm and Nuance can be lost in a simple ML model
- Short or vague answers could be too short to contain enough information for an accurate prediction

Feature: reviewerRating

There are 6 different levels of this variable, pictured below: There is a big skew towards higher

0.0	71
0.4	5244
0.6	9160
0.2	10296
0.8	18719
1.0	67653

Figure 1: reviewerRating

ratings. Reviews with 5, 4, and 1 stars are the most frequent. The reviews of 0 are something to take note of.

Limitations / Considerations:

- Items that have 0-star reviews could give unnecessary noise to the ML model
- with 5-star reviews being much more frequent than other levels, this could introduce bias in the model

Feature: `reviewerDate`

Originally, this date column had multiple different formats for the data points. To fix this inconsistency, we converted everything into a yyyy/mm/dd format. The dates of reviews are sporadic, but the data starts from 2009/02/11 and ends at 2017/10/28. After plotting the counts of reviews by date, there is a steadily increasing trend of reviews as the years pass.

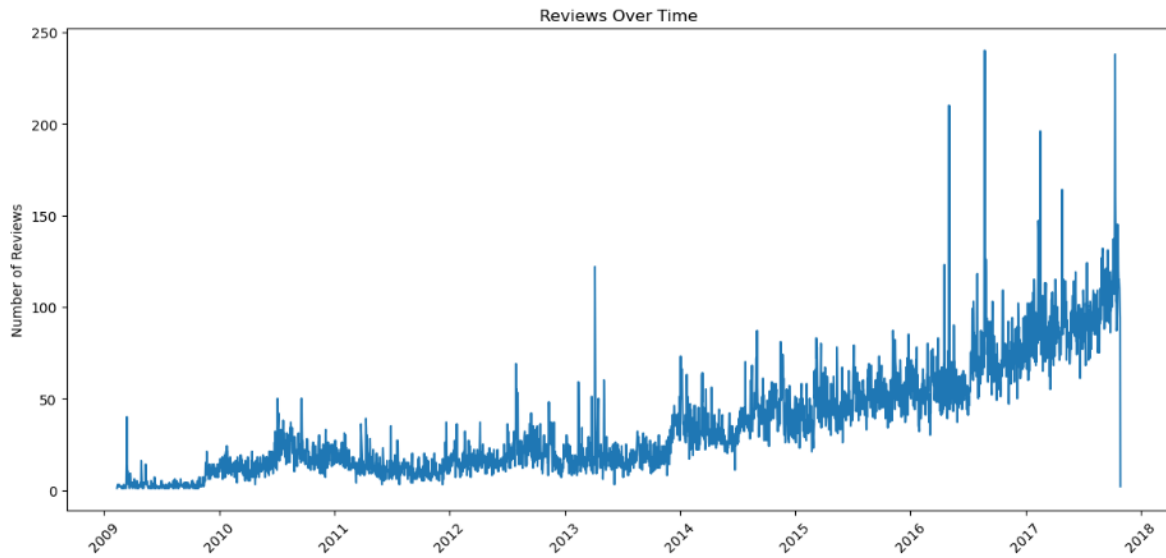


Figure 2: `reviewerDate`

Limitations / Considerations:

- Language evolution could change the meaning of reviews. Words that were once positive might become "neutral" or "negative" as slang or industry standards change.
- Early adopter bias could be introduced, as the first people to buy a product are more likely to rate a product higher

Feature: `textAnalytics`

Most of the values in this column are null (110,656). The rest of the 487 rows that have values in them, and they seem to be on a range from 0.0 to 100.0. Looking at the other data in the row does not give a good indication of what it means.

Limitations / Considerations:

- Lack of values in most columns could make a case for not using this variable.
- Including this variable could cause the ML model to overfit to reviews with present `textAnalytics`.